



Adsorption isotherm, kinetic modeling and mechanism of tetracycline on *Pinus taeda*-derived activated biochar

Hyun Min Jang^a, Seunghyun Yoo^b, Yong-Keun Choi^{a,c}, Sunkyu Park^b, Eunsung Kan^{a,c,*}

^a Texas A&M AgriLife Research Center, 1229 North US Highway 281, Stephenville, TX 76401, USA

^b North Carolina State University, Department of Forest Biomaterials, 2820 Faucette Blvd, Raleigh, NC 27607, USA

^c Office of Sponsored Projects, Tarleton State University, 1333 W. Washington, Stephenville, TX 76401, USA

ARTICLE INFO

Keywords:

Biochar
Tetracycline
Adsorption
Kinetic
Isotherm

ABSTRACT

The objective of this study was to evaluate the adsorption of tetracycline (TC) on the *Pinus taeda*-derived activated biochar (BC). After NaOH activation, the well-developed porous surface structure was observed with a significantly increase in surface area (959.9 m²/g). The kinetic and isotherm studies indicated that hydrogen bonding and π - π interaction on the heterogeneous surface would be the possible mechanisms, while intra-particle diffusion was considered as the major limitation for the adsorption of TC on the activated BC. The maximum adsorption capacity of the activated BC (274.8 mg TC/g BC) was higher than those of various activated BCs from the previous studies while it was similar to those of commercial activated carbons. It indicated that the activated BC had the high potential for TC removal in water.

1. Introduction

Various pharmaceutical antibiotics such as tetracyclines (TCs), sulfonamides, macrolides, and quinolones are widely used for the prevention and treatment of infectious diseases (Yu et al., 2016). Among these antibiotics, TCs are one of the most commonly used antibiotics due to their relatively low costs and effective treatment. Similar to other antibiotics, low metabolic efficiencies of TCs have caused deleterious effects on the human and animal health. According to Daughton and Ternes (1999), about 90% of the ingested TCs was excreted through the urine. In addition, TCs orally applied to pig were rapidly excreted through the urine and feces (Winckler and Grafe, 2001). Previous studies reported that TCs were detected in various water bodies and soil (Halling-Sørensen et al., 1998; Tolls, 2001).

Numerous approaches including biological, chlorination, electrochemical, membrane, ultrasonic cavitation and adsorption process have been used to remove TCs in wastewater and water (Yu et al., 2016). Compared to other technologies, adsorption has distinctive advantages of easy operation, low cost, high removal efficiency at very low concentration, and no toxicities generated from intermediates and by-products (Putra et al., 2009). Recently, biochar (BC) among various adsorbents has received enormous attention due to its advantageous properties including various functional groups and rich aromaticity (Peiris et al., 2017). Since BC is produced from various wastes with lower energy consumption than activated carbon, it can be a cost-

effective adsorbent for removal of TC and other pharmaceuticals in wastewater and water (Peiris et al., 2017). Various BCs derived from crop residue (Jing et al., 2014), sawdust (Zhou et al., 2017), plant (Zhu et al., 2014b), sewage sludge (Yang et al., 2016), and microalgae (Peng et al., 2014) have been applied to removal of TC in wastewater and water. However, the adsorption of TC onto raw (pristine) BC has been limited to low capacities (below 20 mg TC/g BC) owing to mainly the limited surface areas and less developed pore structures (Yang et al., 2016; Zhu et al., 2014b). Recent studies investigated the activation of BC to enhance its pore structures and surface area for increasing its removal efficiency and adsorption capacity (Ahmed, 2017). Up to date the activation processes for BC have employed physical methods (steam and thermal), acids (i.e., H₃PO₄, H₂SO₄, HNO₃ and HCl), alkaline chemicals (KOH and NaOH), activating agents (ZnCl₂, K₂CO₃, FeCl₃, H₂O₂, O₃) and physiochemical activations (Ahmed, 2017; Peiris et al., 2017; Rajapaksha et al., 2016). Zhu et al. (2014a) and Zhu et al. (2014b) reported that the Fe-doped and thermal activation (700 °C for 4 h) drastically increased the surface area of *Salix psammophila*-derived BC (m²/g) significantly from 7.16 to 349 and 316 (m²/g) while increasing adsorption capacity of TC (mg TC/g BC) from 2.71 to 25.44 and to 28.77, respectively. Yang et al. (2016) and Jing et al. (2014) also observed the maximum adsorption of TC onto ferric activated BC (87.87 mg/g) and methanol-modified BC (95.63 mg/g), respectively. In addition, Torres-Pérez et al. (2012) reported the reported the high adsorption capacity of TC (288.3 mg/g) when applied the steam activated

* Corresponding author at: Texas A&M AgriLife Research Center, 1229 North US Highway 281, Stephenville, TX 76401, USA.
E-mail address: eunsung.kan@ag.tamu.edu (E. Kan).

BC with high surface area (821 m²/g).

Considering the wood residues account for 39% of total biomass available in US and pine wood is widely available biomass in the southern part of the US (Thangalazhy-Gopakumar et al., 2010), pine wood-derived BC has the advantages associated with using locally available waste biomass, which significantly reduce the overall carbon footprint of BC production. To our best knowledge, for the first time the present study investigated the detailed characteristics and mechanisms for adsorption of TC onto the *Pinus taeda*-derived activated BC via various adsorption kinetic and isotherm models. Hence, the objectives of this study were to (1) investigate the adsorption characteristics of TC onto the activated BC, (2) determine the kinetic and isotherm models associated with the adsorption of TC onto the activated BC, and (3) figure the underlying mechanisms for the adsorption of TC on the activated BC.

2. Materials and methods

2.1. Reagents

TC (C₂₂H₂₄N₂O₈; purity > 99.0%; CAS Number: 60-54-8) was purchased from Alfa Aesar Co. (Haverhill, Massachusetts, USA). All other chemicals used in this study were analytical grade (purity > 99%) and purchased from Sigma-Aldrich (St. Louis, Missouri, USA). It should be noted that solubility of TC in water is importance to design appropriate adsorption experiments. According to previous research (Caço et al., 2008), the solubility of TC in water at various temperatures were 201 mg/L (at 15.15 °C), 311 mg/L (at 20.15 °C) and 430 mg/L (at 30.15 °C). Based on the solubility of TC in water at ambient conditions 10–100 mg/L of TC were used for the adsorption experiments in this study. Three commercial activated carbon were purchased from Sigma-Aldrich (Darco® G-60 and Norit® GAC) and Calgon Carbon Corp. (Calgon F400) to compare the TC adsorption capacity of these activated carbons with that of the activated BC at the same conditions.

2.2. Preparation of activated BC

As previously described (Park et al., 2013), a 3 g of Debarked loblolly pine (*Pinus taeda*) chips (20–40 mesh sieves and air-dried) was loaded into the preheated quartz-tube furnace (MTI Corporation, Richmond, CA) and then treated at 300 °C for 15 min with continuous nitrogen flow (1 L/min). The first treated sample was designated as “raw BC”. Both KOH and NaOH have been widely used for alkaline activation of BC, while Tseng (2006) suggested that NaOH is more environmentally friendly and less corrosive than KOH. Thus, in this study, BC was activated by an impregnation method using NaOH solution. Briefly, a 3 g of raw BC was soaked in 40 mL of 4 M NaOH solution and incubated at room temperature for 2 h. Afterward, the NaOH-impregnated raw BC was dried overnight at 105 °C and then the dried sample was thermally activated in the tube furnace. The tube furnace was heated from room temperature to 800 °C at 3 °C/min. The sample remained in the furnace for 2 h under an inert atmospheric condition (2 L/min of N₂ flow). The activated sample was washed with 2 L of deionized (DI) water followed by 0.1 M HCl solution (200 mL) wash. Acid washed sample was washed again with DI water until the pH of filtrates was about 7.0. The washed and dried sample was designated as “activated BC”.

2.3. Characterization of BC

Elemental analyzer (PerkinElmer 2400 Series II, MA, USA) was used to determine the Carbon (C), Nitrogen (N) and Hydrogen (H) content in the raw and activated BCs. Oxygen content was calculated by difference based on the ash and C, H, N contents. According to the ASTM standard D7582-15, proximate analysis was conducted and the morphology of each BC samples were observed with a scanning electron microscope

(SEM) (FEI Verios 460, Hillsboro, OR, USA).

Brunauer-Emmett-Teller (BET) surface area and pore-size distribution were evaluated on the basis of the measurements relating to N₂ adsorption with an apparatus at 77 K (Micromeritics Gemini VII 2390p, Norcross, USA). X-ray photoelectron spectroscopy (XPS) analysis was performed using XPS/UPS – SPECS System (SPECS Surface Nano Analysis GmbH, Berlin, Germany) equipped with a monochromatic Mg K α radiation to quantify carbon-oxygen functional groups. All spectra were acquired from a sample area of 1 mm $\times$ 1 mm. The resolution of PHOIBOS 150 photoelectron analyzer is lower than 1 eV.

Energy loss spectroscopy (EELS) was done to analyze sp² content. EELS samples were prepared by using UC7 Ultramicrotome (Leica Microsystems Inc. Buffalo Grove, IL, USA). The raw and activated BCs were impregnated in the resin and sliced into 100 nm thickness. Then, sliced specimen was transferred to the copper TEM grid (Protochips, Raleigh, NC, USA). Electron energy loss spectrum was collected with the electron energy loss spectrometer (Gatan, Pleasanton, CA, USA) attached to probe aberration corrected scanning transmission electron Microscope (FEI Titan 80–300, Hillsboro, OR, USA). Carbon K-edge energy loss spectrum was deconvolved into three Gaussian spectra for the further calculation (Marriott et al., 2014; Zhang et al., 2011).

2.4. Batch adsorption experiments: pH effects, kinetic and isotherm studies

To investigate the effects of initial solution pH on adsorption of TC in water, the batch experiments were conducted at different pH ranging from 3 to 9. The initial solution pH was adjusted by using 0.1 M of NaOH or HCl solution. A weighed amount of BC (0.01 g) was added to 0.1 L of TC solution (100 mg/L) in a glass bottle and agitated at 150 rpm and 20 °C for 7 d. After 7 d, the supernatant was filtrated using a 0.45 μ m syringe filter and then quantified by high performance liquid chromatography (HPLC).

The adsorption kinetics was investigated by adding 0.01 g of the activated BC into 0.1 L of TC solution (20 and 100 mg/L) at pH 6 which was determined as the optimum pH from the above experiments. The solution was then vigorously shaken at 150 rpm for 3 d while the samples were collected at a regular interval.

In addition, the adsorption isotherm experiments were carried out by vigorously stirring 0.1 L of TC solution (10 to 100 mg/L, pH 6) mixed with 0.01 g of the activated BC for 7 d to reach equilibrium. After 7 d, the TC concentration in the supernatant was quantified for estimating the TC concentration in aqueous and solid (BC) phases at equilibrium which were used for development of an appropriate isotherm model.

The pH of zero point charges (pH_{PZC}) of the raw and activated BCs were determined as described by the previous study (Zhu et al., 2014a): (1) 50 mL NaCl solution (0.01 M) was placed in a 100 mL glass bottle to maintain the ion strength of the solution, (2) the pH of solution was adjusted ranged from 3.0 to 10.0 using 0.1 M of NaOH or HCl solution, (3) 0.01 g BC was added to the solution and the bottle was filled with nitrogen gas to minimize the effect of CO₂ in head space of bottle, and then (4) the final pH of solution was measured after 2 d incubation at 20 °C and 150 rpm. The pH_{PZC} from each set were calculated based on the Δ pH (final – initial pH) = 0 (Table 2 and Fig. 1a). It should be noted that all experiments were performed in duplicate while the control experiment was conducted along with all experiments simultaneously to minimized error.

2.5. Adsorption isotherm and kinetic modeling

As shown in Table 1, the six adsorption kinetic models (pseudo-first order, pseudo-second order, Elovich, Two-compartment first-order, Intra-particle diffusion and liquid film diffusion models) and three isothermal models (Langmuir, Freundlich and Temkin models) were used to evaluate the kinetic and isotherm models for adsorption of TC on the activated BC (Zhou et al., 2017).

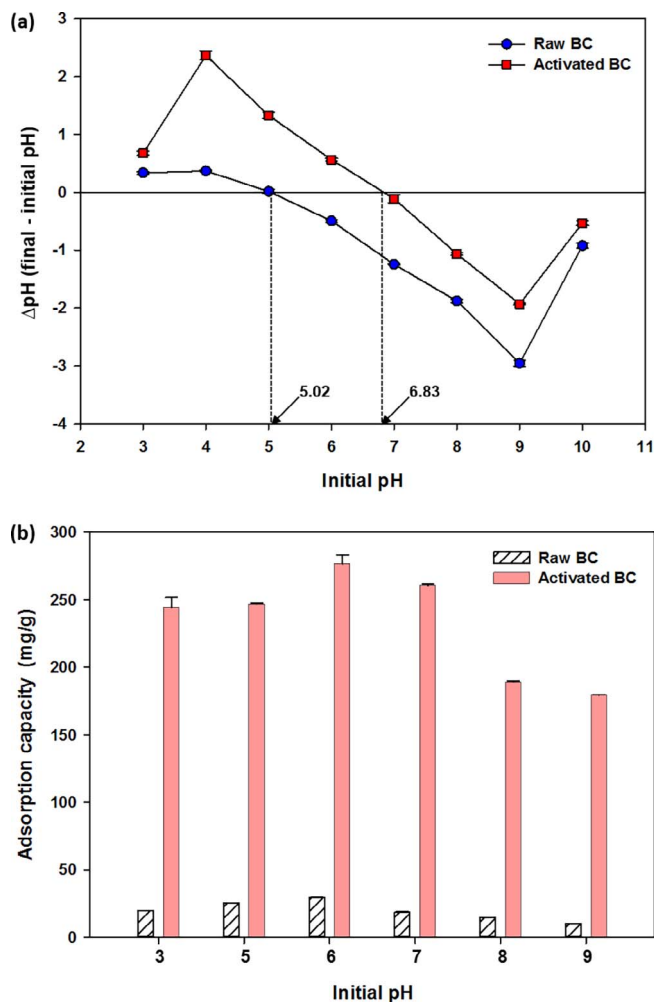


Fig. 1. The pH_{PZC} of the raw and activated BCs (at 20 °C for 2 d) (a) and effect of initial pH on the adsorption of TC on raw and activated BCs (initial TC concentration of 100 mg/L at 20 °C for 7 d) (b).

2.6. Analytical methods

The concentration of TC in aqueous phase was determined by a HPLC (LC-2030C model, SHIMADZU Corp., Torrance, CA, USA) equipped with a C-18 column (Aeris PEPTIDE 3.6 μ m XB-C18, Phenomenex Inc.) and UV–vis detector operating at wavelength of 355 nm. The liquid chromatography column temperature was maintained at 30 °C. The mobile phase was methanol and water with 0.1% formic acid with a linear isocratic ratio of 33:67. All the glass bottles were wrapped with aluminum foils to prevent light induced decomposition during the sorption. Unlike the previous study for TC adsorption onto BC (Wang et al., 2017), the measurable loss of TC was observed from the control experiments (i.e. the experiment without BC). Thus, in this study, the loss of TC in the control experiment sets was compensated to all of the adsorption experiments to minimize the error from the auto-lysis of TC in water.

The adsorption capacity at t time, Q_t (mg/g) was determined according to Eq. (1):

$$Q_t = \frac{(2C_0 - C_t - C_a) \times V}{W} \quad (1)$$

where C_0 and C_t are the TC concentration of initial and at time t (mg/L), respectively, C_a is the TC concentration at time t in control experiment (mg/L), V is the volume of TC solution (L) and W is the g of BC used in study.

The sum of squared error (SSE) was determined according to Eq. (2)

Table 1

Adsorption isotherm and kinetic models used in this study.

	Names	Equations
Isotherm models	Langmuir	$Q_e = \frac{Q_m K_L C_e}{1 + K_L C_e}, R_L = \frac{1}{1 + K_L C_0}$
	Freundlich	$Q_e = K_f C_e^{1/n_f}$
	Temkin	$Q_e = \frac{RT}{bT} \ln(K_T C_e)$
Kinetic models	Pseudo-first order	$Q_t = Q_e (1 - \exp(-K_1 t))$
	Pseudo-second order	$Q_t = \frac{K_2 Q_e^2 t}{1 + K_2 Q_e t}$
	Elovich	$Q_t = \left(\frac{1}{b}\right) \ln ab + \left(\frac{1}{b}\right) \ln t, t_0 = \frac{1}{ab}$
	Intra-particle diffusion	$Q_t = K_i \sqrt{t} + C_i$
	Film diffusion	$\ln(1-F) = -K_{fd} t, F = \frac{Q_t}{Q_e}$
	Two-compartment first order	$\frac{Q_t}{Q_{t=\infty}} = F_{fast} (1 - \exp^{-K_{fast} t}) + F_{slow} (1 - \exp^{-K_{slow} t}), F_{fast} + F_{slow} = 1$

Q_m = maximum adsorption capacity (mg/g), K_L = Langmuir constant, C_e = equilibrium concentration (mg/L), R_L = separation constant, K_f and n_f = Freundlich constant, R = universal gas constant (8.314 J/mol), T = temperature in terms of Kelvin, b_T = Temkin constant, K_T = equilibrium bond constant related to the maximum energy of bond, t = time (min), Q_e = adsorption capacity (mg/g) at equilibrium time, Q_t = adsorption capacity (mg/g) at time t (min), K_1 = rate constant of pseudo-first order, K_2 = rate constant of pseudo-second order, a = rate constant of chemisorption, b = constant of the surface coverage, K_i = intra-particle diffusion rate constant (mg min^{0.5}/g), C_i = a constant (mg/g), K_{fd} = adsorption rate constant, F_{fast} = mass fraction of fast, F_{slow} = mass fraction of slow, K_{fast} = first order rate constant for transfer into fast (h⁻¹), K_{slow} = first order rate constant for transfer into slow (h⁻¹), h = hour.

(Zhou et al., 2017):

$$SSE = \sqrt{\frac{\sum (Q_e - Q_c)^2}{N}} \quad (2)$$

where Q_e and Q_c are the observed and calculate value of Q , and N is the number of measurements.

The R^2 was determined by Eq. (3):

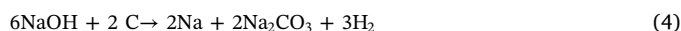
$$1 - \frac{\sum (\hat{Y} - \bar{Y})^2}{\sum (\hat{Y} - \bar{Y})^2} \quad (3)$$

where $\hat{Y}$ is the observed value, $\bar{Y}$ is the fitted value and $\bar{Y}$ is the mean of observed values.

3. Results and discussion

3.1. Characterizations of raw and activated BCs

The chemical activation enhanced the adsorption capacity of BC by creating additional adsorption sites, which was a result of increased surface areas and interactions with specific surface functional groups. In general, NaOH can accelerate the dissolution of condensed organic matter (e.g. lignin and celluloses) during the impregnation step and it promoted a subsequent chemical reaction during the thermal activation stage (Rajapaksha et al., 2016). According to the literature (Martins et al., 2015), predominant reaction stoichiometry between NaOH and carbon during the chemical activation is given as (Eq. (4)):



Well-developed micropore structure can be achieved by evolution of CO and CO₂ which are decomposition products of Na₂CO₃ as well as hydrogen with strong reducing power. In addition, interaction between alkali compounds and carbon promotes the formation of high specific surface area as described previously (Ota et al., 1990).

SEM micrographs of raw and activated BCs were illustrated in the Supplementary data. The surface of raw BC presented a flat structure

Table 2
Physical, chemical and textual characteristics of raw and activated BCs.

	Elemental analysis (wt %)						Proximate analysis (% dry basis)			
	C	H	N	O ^a	O/C	H/C	Volatile compounds	Fixed carbon	Ash	pH _{PZC} ^b
Raw biochar	56.4	5.9	0.1	37.6	0.67	0.10	87.4	11.0	1.5	5.08
Activated biochar	85.8	0.5	0.3	13.5	0.16	5.8×10^{-3}	10.1	86.8	3.1	6.83
	Textural analysis			Oxygen-containing functional group (At.%)						
	BET surface area (m ² /g)	Volume of pores (cm ³ /g)	Average of pore width (nm)	C–C, C=C, C–H (284.6 eV)	C–O–H, C–O–C (286.2 eV)	C=O (287.6 eV)	O=C–O (289.1 eV)			
Raw biochar	1.4	0.008	24.4	82.6	8.4	7.5	1.4			
Activated biochar	959.9	0.4	4.4	77.8	11.1	7.4	3.7			

^a Calculated by difference.

^b The pH of zero point charges.

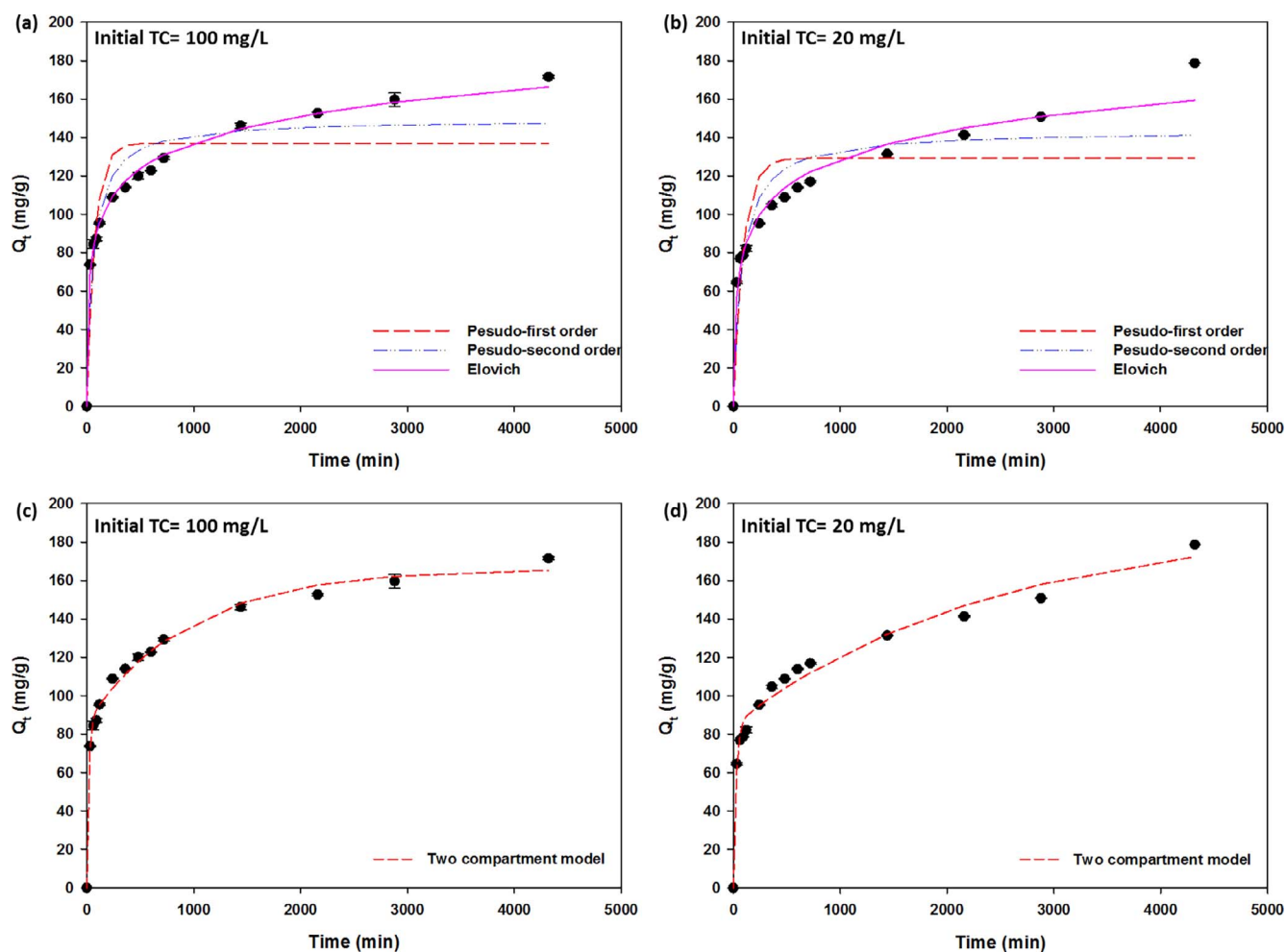


Fig. 2. Kinetic data of TC adsorption on activated BC at 20 °C and pH = 6 under two different initial TC concentration (20 and 100 mg/L).

and there is no clear evidence of porous structure development. In contrast to the raw BC, the apparent porous structure was observed on the surface of activated BC, indicating the well-developed pore structure after the activation. These results are highly consistent with the BET surface areas of raw and activated BCs (Table 2). As shown in Table 2, the activated BC had a large BET surface area (959.9 m²/g) and high pore volume (0.4 cm³/g) compared to raw BC (1.4 m²/g of BET surface area and 0.008 cm³/g of total pore volume). It indicated that the alkaline activation of raw BC resulted in the increase in the BET surface

area (about 659-fold increase). Similarly, the previous study (Chia et al., 2015) observed the alkaline activation of BC led to the extremely high surface area of BC. In this study, significant decrease in average of pore width (from 24.4 to 4.4 nm) was observed after activation. According to Martins et al. (2015), TC species have sizes smaller than 1.27 nm, suggesting that it is reasonable to expect that the activated BC could be suitable for TC adsorption with sufficient pore size to allow penetration of TC into pores.

The elemental compositions of the raw and activated BCs are also

Table 3
Kinetic parameters of TC adsorption on activated BC.

	20 ^a	100 ^a
Qe ^b	178.69	171.51
<i>Pseudo-first order</i>		
Qe ^c	129.29	136.94
K ₁	0.0101	0.0131
SSE	20.8759	18.3534
R ²	0.7509	0.8097
<i>Pseudo-second order</i>		
Qe ^c	143.73	149.45
K ₂	8.87 × 10 ⁻⁵	0.0001
SSE	15.3808	12.6044
R ²	0.8648	0.9103
<i>Elovich</i>		
Qe ^c	178.69	166.28
a	10.3707	20.9736
b	0.0481	0.0507
t ₀	2.0047	0.9404
SSE	6.7066	2.9049
R ²	0.9743	0.9952
<i>Two-compartment first-order</i>		
Qe ^c	190.09	166.21
F _{fast}	0.4474	0.5182
F _{slow}	0.5526	0.4818
K _{fast}	2.3363	3.2672
K _{slow}	0.0248	0.0624
SSE	5.1522	3.4126
R ²	0.9848	0.9934

^a Initial concentration of TC (mg/L).

^b Observed value.

^c Calculated value.

summarized in Table 2. After activation, the C content in BC increased up to 85.8%, while the O and H contents decreased significantly from 37.6 to 13.5% and from 5.9 to 0.5%, respectively. Interestingly, these behaviors of O and H contents in BC contradicts the previous study (Liu et al., 2012) which reported the increased O and H contents in rice husk BC after alkaline (KOH) activation. Comparing the low activation temperature (60–70 °C) of previous study (Liu et al., 2012) and the high activation temperature (800 °C) of present study, this discrepancy can be explained by decomposition of O and H under high temperature as described reported by others (Wang et al., 2017). The ratios of O/C and H/C exhibited a similar trend. After activation, the ratio O/C (polarity) and H/C (aromaticity) decreased from 0.67 and 0.10 to 0.16 and 5.8×10^{-3} , respectively. This result is consistent with data from Park et al. (2013) which reported a growing aromatic structure using ¹³C DP/NMR analysis. Taken together, these changes in elemental composition indicated that the NaOH activation promoted the hydrophobicity with increased sp² aromatic carbons content from 58.0 to 73.7% (data not shown).

To further understand the nature of surface functionalities, C 1s photoelectrons of XPS were investigated (Table 2). The high resolution C 1(s) photoelectron spectrum for the raw and activated BCs included four common signals, which can be attributed to the aliphatic/aromatic carbon groups (C–C, C=C and C–H, 284.6 eV), hydroxyl group (C–O–H and C–O–C, 286.2 eV), carbonyl group (C=O, 287.6 eV), and carboxylic/ester/lactone group (O=C–O, 289.1 eV) (Liu et al., 2012; Zhu et al., 2014b). Similar to the results reported by Zhu et al. (2014b), the significant decomposition of volatile compounds was observed from 87.4 to 10.1%. Although decomposition of volatile compounds caused decrease in oxygen content by releasing CO₂ and CO, the oxygen-containing groups of activated BC slightly increased resulted from additional generation of carbon-oxygen functional groups.

3.2. Effect of initial solution pH on adsorption of TC onto the raw and activated BCs

The solution pH often influences adsorption of TC on BC by altering properties of both the TC and BC. TC is an amphoteric molecule with three values of pK_a (3.3, 7.7 and 9.7) and has differently changed functional groups at different pH ranges. Their predominant species are H₄TC⁺ at pH < 3.4, H₃TC at 3.4 < pH < 7.6, H₂TC[−] at 7.6 < pH < 9.7, and HTC^{2−} at pH > 9.7, respectively (Zhou et al., 2017; Zhu et al., 2014a). As shown in Fig. 1a and Table 2, the pH_{PZC} values of raw and activated BCs were 5.02 and 6.83, respectively. Thus, the surface charge of BC was protonated (i.e. positive charge) when the solution pH was lower than their pH_{PZC}, otherwise it was the negative. At the pH ranges (3–3.4, 7.6–9), TC molecules (H₄TC⁺ and H₂TC[−]) have repulsive interactions with positive and negative charges of the activated BC. On the other hand, at pH between 3.4 (i.e., pK_{a1}) and 7.6 (i.e., pK_{a2}), TC molecules (i.e. H₃TC) carry no net electrical charge while the activated BC had slightly negatively or positively charged, which minimize the electrostatic attraction or repulsion. For this nature, at pH ranged from 4.0 to 7.0 has no significant effect on the TC adsorption capacity in this study (Fig. 1b). It should be noted that considerable amount of TC was adsorbed on the activated BC at pH 3, although electrostatic repulsion (i.e. positive – positive repulsion) occurred. This result suggests the adsorption of TC onto the activated BC at pH 3 would also possess the different interactions between TC and activated BC (i.e., hydrogen bonding and π-π electron donor–acceptor (π-π EDA) interactions) (Zhu et al., 2014a). At higher pH (> 7.0), a progressive decrease in the adsorption capacity on the activated BC was observed. This could be attributed to a strong electrostatic repulsion between the negatively charged TC and surface of BC with weak π-π EDA interaction (Zhou et al., 2017). Further batch adsorption experiments were conducted at pH 6 observed maximum adsorption capacity of TC (approximately 277 mg/g).

Similar to the activated BC, the raw BC showed the highest adsorption of TC at pH 6 followed by pH 5 and 7 although the adsorption capacities of the raw BC were much lower than those of the activated BC (Fig. 1b).

3.3. Adsorption kinetics

The kinetic parameters are important information to understand adsorption mechanisms and rate-limiting step which are useful for designing and modeling of commercial applications. The dynamic adsorption of TC onto the activated BC was conducted with two initial concentrations of TC (20 and 100 mg/L) at pH 6 and 20 °C (Fig. 2 and Table 3). It is interesting to note that the curves from two initial concentrations of TC exhibit the similar trend. The adsorbed TC onto the activated BC increased rapidly at the early stage of the TC adsorption, while slower adsorption rate was obtained as a consequence of the decrease of active adsorption sites. This phenomenon is frequently observed during adsorption of TC on BC (Peng et al., 2014; Zhou et al., 2017). In this study, the pseudo-first order, pseudo-second order and Elovich kinetic models (Table 1) were applied to the adsorption kinetic data for understanding possible mechanism associated with the adsorption of TC onto the activated BC. According to recent review (Peiris et al., 2017), pseudo-second order was the major fitted model for TC adsorption on the BC while some studies reported pseudo-first order fitted experimental data. Interestingly, the models of the pseudo-first order and pseudo-second order were not suitable to described the interaction between TC and activated BC due to the relatively low R² (0.75–0.91) and high SSE (12.60–20.88) (Fig. 2 and Table 3). However, the Elovich model was defined as being the well-fitted (i.e. highest R² (0.97–0.99) and lowest SSE (2.90–6.71)) model to explain the kinetic data. Taking into account the assumption of this model, surface of activated BC was energetically heterogeneous and chemisorptive interaction occurred on surface (Martins et al., 2015; Zhou et al., 2017; Zhu

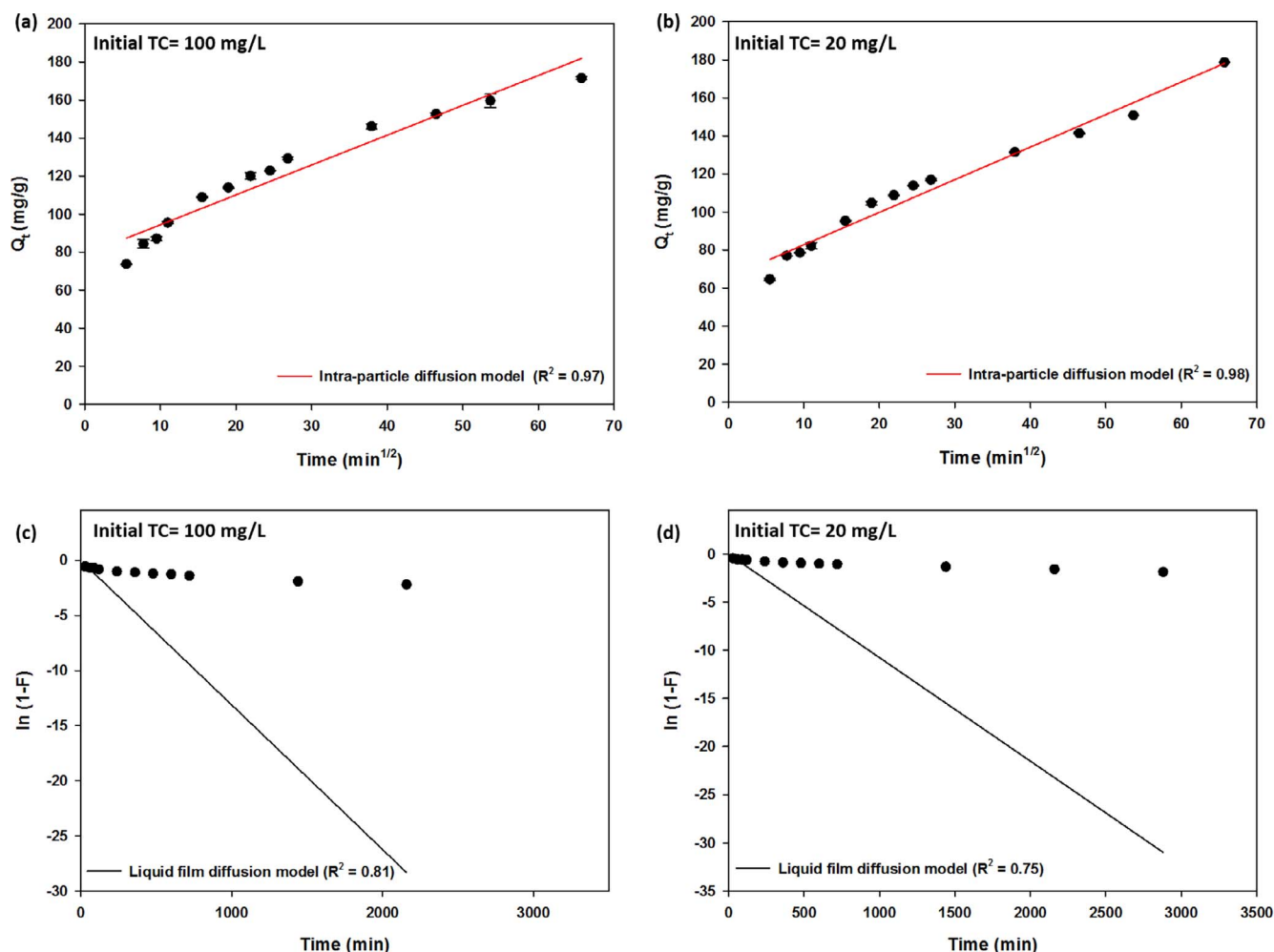
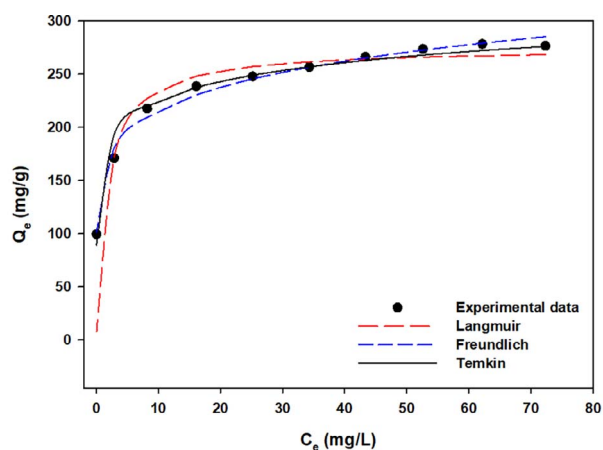


Fig. 3. Kinetics of TC adsorption on activated BC by fitting the intra-particle model (a and b), (b) and film diffusion model (c and d).



Langmuir			Freundlich			Temkin		
K_L	Q_m	R^2	K_f	n_f	R^2	b_T	K_T	R^2
0.58	274.81	0.69	155.45	7.05	0.99	94.86	648.06	0.98

Fig. 4. Adsorption isotherms of TC by fitting the Langmuir, Freundlich and Temkin isotherm models. More detailed information of isotherm models is summarized in Table 1.

et al., 2014a). Similar results were obtained by Martins et al. (2015) who reported kinetics of TC adsorption on NaOH-activated BC from macadamia nut shells. In addition, two-compartment first-order which considered the adsorption TC on activated BC as a two-domain process

(Zhou et al., 2017) was analyzed, presenting high R^2 (0.98–0.99) and low SSE (3.41–5.15) (Fig. 2 and Table 3). The higher K_{fast} (2.34–3.27) than K_{slow} (0.02–0.06) suggested that the fast sorption stage resulted from hydrogen bonding and π - π EDA may be the possible mechanisms for the adsorption of TC on activated BC.

Adsorption kinetics typically involves four steps as follows (Ahmed et al., 2015): (1) bulk transport, (2) film transport, (3) intra-particle transport and (4) adsorption on the adsorbent. In order to understand major limitations in the adsorption of TC onto the activated BC, the intra-particle diffusion and film diffusion models were used in this study. As shown in Fig. 3, the plots (Q_t VS $t^{1/2}$) showed high R^2 (0.97–0.98) while the plots ($\ln(1 - F)$ VS t) were not fitted to experimental data with low R^2 (0.75–0.81). This result indicated that intra-particle diffusion would be the major limitation for the adsorption of TC on the activated BC. However, at the early stage of adsorption, the curves for intra-particle diffusion model did not pass through the origin, indicating the involvement of film diffusion in the mechanism of adsorption (Martins et al., 2015; Zhou et al., 2017; Zhu et al., 2014a).

3.4. Adsorption isotherms

Three widely used isotherm models (i.e. Langmuir, Freundlich and Temkin) (Ahmed, 2017) were used to be fitted to experiment data (Fig. 4). Langmuir model describes the monolayered adsorbents on the homogeneous surface (Gupta and Bhattacharyya, 2011). In several studies of TC adsorption, Langmuir model was best fitted (Peiris et al., 2017). However, in this study, Langmuir model was not fitted

Table 4Maximum adsorption capacity (Q_m) of TC and the surface area of various carbonaceous materials (CM).^a

Raw material	Surface area (m ² /g)	Q_m (mg/g)	Adsorption conditions	References
Biochar (Rice straw)	27.66	13.85 ^c	$C_0 = 0.5\text{--}32$ mg/L, 35 °C, 3 g/L of CM	Wang et al. (2017)
Hydro-char (<i>Salix psammophila</i>)	7.16	2.71 ^c	$C_0 = 5\text{--}50$ mg/L, 30 °C, 1 g/L of CM	Zhu et al. (2014b)
Thermal activated hydro-char (<i>Salix psammophila</i>)	316	28.77 ^c		
Ferric activated hydro-char (<i>Salix psammophila</i>)	349	25.44 ^c	$C_0 = 5\text{--}80$ mg/L, 30 °C	Zhu et al. (2014a)
Methanol activated biochar (Rice husk)	65.97	95.63 ^b	$C_0 = 100$ mg/L, 30 °C, 1 g/L of CM	Jing et al. (2014)
Biochar (Sewage sludge)	2.33	15.23 ^c	$C_0 = 5\text{--}150$ mg/L, 45 °C	Yang et al. (2016)
Ferric activated biochar (sewage sludge)	126.86	87.87 ^c		
Steam activated biochar (Beet pulp)	821	288.3 ^c	$C_0 = 15\text{--}300$ mg/L	Torres-Pérez et al. (2012)
Steam activated biochar (Peanut shell)	829	28.0 ^c		
Commercial activated carbon (Steam activated coconut shell)	1138	133.1 ^c		
Commercial activated carbon (H ₃ PO ₄ activated wood)	1515	816.7 ^c		
Multi walled carbon nanotubes	207	192.7 ^c	$C_0 = 100$ mg/L, 30 °C, 7.5–500 mg of CM	Álvarez-Torrellas et al. (2016)
Biochar (<i>Pinus taeda</i>)	1.4	29.42 ^b	$C_0 = 10\text{--}100$ mg/L, pH = 6, 20 °C, 0.1 g/L of CM	In this study
NaOH activated biochar (<i>Pinus taeda</i>)	959.9	274.81 ^c		
Commercial activated carbon (Calgon F400)	816.3 ^d	201.75 ^b		
Commercial activated carbon (Darco® G-60)	933 ^e	290.09 ^b		
Commercial activated carbon (Norit® GAC)	1200 ^f	211.58 ^b		

^a Based on the literature (Caço et al., 2008), references used higher C_0 than 300 mg/L were not included.^b Derived from experimental data.^c Derived from isotherm model data.^d Gauden et al. (2006).^e Oliveira et al. (2002).^f Gicheva and Yordanov (2013).

($R^2 = 0.69$ and $SSE = 29.94$) with experimental data. The lower separation factor (R_L) than 1 (i.e. $K_L > 0$) suggests the TC adsorption on activated BC is favorable in the concentration range used in this study (Martins et al., 2015; Zhou et al., 2017). As shown in Fig. 4, Freundlich model presented R^2 of 0.99 and SSE of 5.60. The well-fitted this model indicated that the adsorption would take place with a heterogeneous surface (Günay et al., 2007). The $1/n_f$ value (0.14) defined as heterogeneity factor presents high heterogeneity degree (Junior et al., 2014). This result is consistent with the kinetic studies (i.e. well-fitted with Elovich model) (Fig. 2). Furthermore, Temkin model showed high R^2 (0.98) and low SSE (8.39) with b_T value (defined as variation of adsorption energy) of 94.86. It can be seen that the TC adsorption on the activated BC surface take place exothermically ($b_T > 1$) resulted from electrostatic interaction and heterogeneous pores (Hadi et al., 2010; Martins et al., 2015).

Table 4 lists the maximum adsorption capacities for TC and surface areas of carbonaceous materials. The Q_m of the activated BC (274.81 mg/g) for TC from this study was higher than those of various activated BCs (ranged from 28.77 to 95.63 mg/g) (Jing et al., 2014; Yang et al., 2016; Zhu et al., 2014a; Zhu et al., 2014b) while it was comparable to that of the steam activated BC (288.3 mg/g) (Torres-Pérez et al. (2012)).

To evaluate feasibility of the activated BC in this study, the values of Q_m from three commercial ACs (Calgon F400, Darco® G-60 and Norit® GAC) were experimentally obtained under the same conditions used for the activated BC in this study (Table 4). Interestingly, the Q_m of the activated BC was higher than those of Calgon F400 (201.75 mg/g) and Norit® GAC (211.58 mg/g), whereas Darco® G-60 showed slightly higher value of Q_m (290.09 mg/g) than that of activated BC. According to recent study (Torres-Pérez et al., 2012), proportion of mesoporous volumes in total pore volume can be directly related to the TC adsorption capacity. For instance, the well fitted curves were observed between the mesoporous volumes and K_f (Freundlich constant). Therefore, it is reasonable to explain that the activated BC in this study possessed the high surface area (959.9 m²/g) with balanced porous structure (i.e. high mesoporous volume, 0.13 cm³/g) via the chemical activation with NaOH resulting in high TC adsorption capacity. However, it is important to clarify that this study was conducted with simple matrix solution (i.e. deionized water), while the adsorption capacity with the complex matrix still unclear due to their characteristics (i.e.

amphoteric molecule). Therefore, as further studies effects of competitive contaminants in water and wastewater (i.e. multiple metal ions, organic contaminants) at various conditions (e.g., pH and temperature) will be investigated for practical application. Another point to take into consideration in future studies is possible regeneration of the activated BC which is a critical aspect in the cost effectiveness (Peiris et al., 2017).

4. Conclusions

The *Pinus taeda*-derived activated BC showed high adsorption ($Q_m = 274.8$ mg/g) for tetracycline (TC) and surface area (959.9 m²/g). The adsorption capacity of the activated BC was higher than those of various activated BCs from the previous studies while it was similar to those of commercial activated carbons. Elovich kinetic and Freundlich isotherm models were found to be the well fitted ones indicating that hydrogen bonding and π - π interaction on the heterogeneous surface would be possible mechanisms while intra-particle diffusion would be the major limitation for the adsorption of TC on the activated BC.

Acknowledgements

This work was supported by Texas A&M University Chancellor Research Initiative Fund and the US-DOE Office of Energy Efficiency and Renewable Energy (Award Number DE-EE0006639).

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.biortech.2018.03.013>.

References

- Ahmed, M.B., Zhou, J.L., Ngo, H.H., Guo, W., 2015. Adsorptive removal of antibiotics from water and wastewater: progress and challenges. *Sci. Total Environ.* 532, 112–126.
- Ahmed, M.J., 2017. Adsorption of quinolone, tetracycline, and penicillin antibiotics from aqueous solution using activated carbons. *Environ. Toxicol. Pharmacol.* 50, 1–10.
- Álvarez-Torrellas, S., Rodríguez, A., Ovejero, G., García, J., 2016. Comparative adsorption performance of ibuprofen and tetracycline from aqueous solution by carbonaceous materials. *Chem. Eng. J.* 283, 936–947.
- Caço, A.I., Varanda, F., Pratas de Melo, M.J., Dias, A.M., Dohrn, R., Marrucho, I.M., 2008.

- Solubility of antibiotics in different solvents. Part II. Non-hydrochloride forms of tetracycline and ciprofloxacin. *Ind. Eng. Chem. Res.* 47 (21), 8083–8089.
- Chia, C.H., Downie, A., Munroe, P., 2015. Characteristics of biochar: physical and structural properties. *Biochar for environmental management: science and technology*. Earthscan Books Ltd, London, 89–109.
- Daughton, C.G., Ternes, T.A., 1999. Pharmaceuticals and personal care products in the environment: agents of subtle change? *Environ. Health Perspect.* 107, 907–938.
- Gauden, P.A., Szmeczig-Gauden, E., Rychlicki, G., Duber, S., Garbacz, J.K., Buczkowski, R., 2006. Changes of the porous structure of activated carbons applied in a filter bed pilot operation. *J. Colloid Interface Sci.* 295 (2), 327–347.
- Gicheva, G., Yordanov, G., 2013. Removal of citrate-coated silver nanoparticles from aqueous dispersions by using activated carbon. *Colloids Surf., A* 431, 51–59.
- Günay, A., Arslankaya, E., Tosun, I., 2007. Lead removal from aqueous solution by natural and pretreated clinoptilolite: adsorption equilibrium and kinetics. *J. Hazard. Mater.* 146 (1), 362–371.
- Gupta, S.S., Bhattacharyya, K.G., 2011. Kinetics of adsorption of metal ions on inorganic materials: a review. *Adv. Colloid Interface Sci.* 162 (1), 39–58.
- Hadi, M., Samarghandi, M.R., McKay, G., 2010. Equilibrium two-parameter isotherms of acid dyes sorption by activated carbons: study of residual errors. *Chem. Eng. J.* 160 (2), 408–416.
- Halling-Sørensen, B., Nors Nielsen, S., Lanzky, P.F., Ingerslev, F., Holten Lützhøft, H.C., Jørgensen, S.E., 1998. Occurrence, fate and effects of pharmaceutical substances in the environment—a review. *Chemosphere* 36 (2), 357–393.
- Jing, X.-R., Wang, Y.-Y., Liu, W.-J., Wang, Y.-K., Jiang, H., 2014. Enhanced adsorption performance of tetracycline in aqueous solutions by methanol-modified biochar. *Chem. Eng. J.* 248, 168–174.
- Junior, O.P., Cazetta, A.L., Gomes, R.C., Barizão, É.O., Souza, I.P., Martins, A.C., Asefa, T., Almeida, V.C., 2014. Synthesis of ZnCl₂-activated carbon from macadamia nut endocarp (*Macadamia integrifolia*) by microwave-assisted pyrolysis: optimization using RSM and methylene blue adsorption. *J. Anal. Appl. Pyrol.* 105, 166–176.
- Liu, P., Liu, W.-J., Jiang, H., Chen, J.-J., Li, W.-W., Yu, H.-Q., 2012. Modification of biochar derived from fast pyrolysis of biomass and its application in removal of tetracycline from aqueous solution. *Bioresour. Technol.* 121, 235–240.
- Marriott, A.S., Hunt, A.J., Bergstrom, E., Wilson, K., Budarin, V.L., Thomas-Oates, J., Clark, J.H., Brydson, R., 2014. Investigating the structure of biomass-derived non-graphitizing mesoporous carbons by electron energy loss spectroscopy in the transmission electron microscope and X-ray photoelectron spectroscopy. *Carbon* 67, 514–524.
- Martins, A.C., Pezoti, O., Cazetta, A.L., Bedin, K.C., Yamazaki, D.A.S., Bandoch, G.F.G., Asefa, T., Visentainer, J.V., Almeida, V.C., 2015. Removal of tetracycline by NaOH-activated carbon produced from macadamia nut shells: kinetic and equilibrium studies. *Chem. Eng. J.* 260, 291–299.
- Oliveira, L.C.A., Rios, R.V.R.A., Fabris, J.D., Garg, V., Sapag, K., Lago, R.M., 2002. Activated carbon/iron oxide magnetic composites for the adsorption of contaminants in water. *Carbon* 40 (12), 2177–2183.
- Otawa, T., Yamada, M., Tanibata, R., Kawakami, M., 1990. Preparation, pore analysis and adsorption behavior of high surface area active carbon from coconut shell. In: *Proceeding of the International Symposium on Gas Separation Technology*, Antwerp (Belgium), Elsevier, pp. 263–270.
- Park, J., Hung, I., Gan, Z., Rojas, O.J., Lim, K.H., Park, S., 2013. Activated carbon from biochar: influence of its physicochemical properties on the sorption characteristics of phenanthrene. *Bioresour. Technol.* 149, 383–389.
- Peiris, C., Gunatilake, S.R., Mlsna, T.E., Mohan, D., Vithanage, M., 2017. Biochar based removal of antibiotic sulfonamides and tetracyclines in aquatic environments: a critical review. *Bioresour. Technol.* 246, 150–159.
- Peng, L., Ren, Y., Gu, J., Qin, P., Zeng, Q., Shao, J., Lei, M., Chai, L., 2014. Iron improving bio-char derived from microalgae on removal of tetracycline from aqueous system. *Environ. Sci. Pollut. Res.* 21 (12), 7631–7640.
- Putra, E.K., Pranowo, R., Sunarso, J., Indraswati, N., Ismadji, S., 2009. Performance of activated carbon and bentonite for adsorption of amoxicillin from wastewater: mechanisms, isotherms and kinetics. *Water Res.* 43 (9), 2419–2430.
- Rajapaksha, A.U., Chen, S.S., Tsang, D.C., Zhang, M., Vithanage, M., Mandal, S., Gao, B., Bolan, N.S., Ok, Y.S., 2016. Engineered/designer, biochar for contaminant removal/immobilization from soil and water: potential and implication of biochar modification. *Chemosphere* 148 (27), 6e291.
- Thangalazhy-Gopakumar, S., Adhikari, S., Ravindran, H., Gupta, R.B., Fasina, O., Tu, M., Fernando, S.D., 2010. Physicochemical properties of bio-oil produced at various temperatures from pine wood using an auger reactor. *Bioresour. Technology* 101 (21), 8389–8395.
- Tolls, J., 2001. Sorption of veterinary pharmaceuticals in soils: a review. *Environ. Sci. Technol.* 35 (17), 3397–3406.
- Torres-Pérez, J., Gérente, C., Andrès, Y., 2012. Sustainable activated carbons from agricultural residues dedicated to antibiotic removal by adsorption. *Chin. J. Chem. Eng.* 20 (3), 524–529.
- Tseng, R.-L., 2006. Mesopore control of high surface area NaOH-activated carbon. *J. Colloid Interface Sci.* 303 (2), 494–502.
- Wang, H., Chu, Y., Fang, C., Huang, F., Song, Y., Xue, X., 2017. Sorption of tetracycline on biochar derived from rice straw under different temperatures. *PLoS One* 12 (8), e0182776.
- Winckler, C., Grafe, A., 2001. Use of veterinary drugs in intensive animal production. *J. Soils Sediments* 1 (2), 66.
- Yang, X., Xu, G., Yu, H., Zhang, Z., 2016. Preparation of ferric-activated sludge-based adsorbent from biological sludge for tetracycline removal. *Bioresour. Technol.* 211, 566–573.
- Yu, F., Li, Y., Han, S., Ma, J., 2016. Adsorptive removal of antibiotics from aqueous solution using carbon materials. *Chemosphere* 153, 365–385.
- Zhang, Z.L., Brydson, R., Aslam, Z., Reddy, S., Brown, A., Westwood, A., Rand, B., 2011. Investigating the structure of non-graphitising carbons using electron energy loss spectroscopy in the transmission electron microscope. *Carbon* 49 (15), 5049–5063.
- Zhou, Y., Liu, X., Xiang, Y., Wang, P., Zhang, J., Zhang, F., Wei, J., Luo, L., Lei, M., Tang, L., 2017. Modification of biochar derived from sawdust and its application in removal of tetracycline and copper from aqueous solution: Adsorption mechanism and modelling. *Bioresour. Technol.* 245, 266–273.
- Zhu, X., Liu, Y., Qian, F., Zhou, C., Zhang, S., Chen, J., 2014a. Preparation of magnetic porous carbon from waste hydrochar by simultaneous activation and magnetization for tetracycline removal. *Bioresour. Technol.* 154, 209–214.
- Zhu, X., Liu, Y., Zhou, C., Luo, G., Zhang, S., Chen, J., 2014b. A novel porous carbon derived from hydrothermal carbon for efficient adsorption of tetracycline. *Carbon* 77, 627–636.